

Next.js + Sanity Marketing Site

CMS content resilience, localisation and Draft Mode QA
on Sanity's official Next.js starter

SYSTEM UNDER TEST	Self-hosted copy of sanity-io/sanity-template-nextjs-clean, unmodified, seeded with 17 adversarial QA documents
STACK	Next.js 16.2.7 (App Router, production build), React 19.2.7, next-sanity 13.0.8, Sanity Studio 5.31.1
SCOPE	CMS content resilience, localisation, accessibility and Draft Mode across six browser and device profiles
DELIVERABLES	This report, the seed script and a Playwright + TypeScript regression suite with known bugs encoded as expected failures

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DATE
7 October 2026

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1 Executive summary

Release assessment: BLOCK before production

CMS-01 is the primary release risk: unknown routes return HTTP 200 and expose a public Studio onboarding prompt. The remaining findings show how realistic CMS edge cases can break layout, localisation, accessibility, media handling, routing and sitemap integrity.

Draft Mode, output escaping, deleted-author handling and unknown post routes behaved correctly. Every confirmed defect is encoded in Playwright and was independently verified against screenshots, DOM state or HTTP output.

14

confirmed findings

+ 1 emulation-only
observation awaiting a real
device

Findings by severity

Critical		0
High		1
Medium		9
Low		4

Top issues

CMS-01	HIGH	Page routing — unknown URLs return HTTP 200 with a public “Create Page” Studio prompt (soft 404)
CMS-03	MEDIUM	Layout — unbreakable words and pasted URLs cause horizontal scrolling
CMS-04	MEDIUM	Localisation — Arabic (RTL) content renders left-to-right, with misordered punctuation and numbers

Test run

318 test executions	138 normal passes	93 expected failures	87 skipped by design	0 unexpected outcomes
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Expected failures encode confirmed defects. Total test time was approximately 3 hours including setup; no unexpected outcomes were recorded.

Coverage

Desktop Chrome

Desktop Safari (WebKit)

Desktop Firefox

iPhone 14

Pixel 7

iPad (gen 7)

2 Scope and methodology

ITEM	DETAIL
Template	<code>sanity-io/sanity-template-nextjs-clean</code> (via <code>create-sanity</code> 6.0.51), unmodified
Stack	Next.js 16.2.7 (App Router; production build with <code>next build</code> & <code>next start</code>), React 19.2.7, next-sanity 13.0.8, Sanity Studio 5.31.1
CMS	Free Sanity project <code>isdq5nsx</code> , dataset <code>production</code> : the template's sample data plus 17 seeded QA documents
Draft and preview	Draft Mode enabled exactly as the Presentation tool does it: a preview-URL secret, then <code>/api/draft-mode/enable</code>
Out of scope	Studio UI usability, performance and load, and security testing beyond output-escaping checks

Adversarial CMS content (seed/build-seed.mjs)

- A 230-character headline; a single unbreakable 120-character word; a pasted 200-character campaign URL
- A post with every optional field empty: no excerpt, image, author or date, and an empty body (`[]`)
- A cover image without alt text, and a cover image whose asset was deleted
- An author reference to a deleted person, and a CTA linking to a deleted page
- Arabic RTL content mixing Arabic and English, numbers and percentages
- HTML-like strings (`<script>` , `<img onerror>` , `&`), emoji with ZWJ sequences, and flags
- A 40-paragraph body; two published posts sharing one slug; a published post without a slug; a page without a slug
- A draft-only post, and a published post with unpublished changes

Methodology

Content was written through the API, which bypasses Studio-only validation in the same way that imports, migrations, scripts and drafts can. The site ran as a production build and was exercised with a six-profile Playwright matrix, axe-core accessibility checks and direct HTTP checks.

Every reported finding was independently re-run and verified using screenshots, DOM state or HTTP output. Emulation-only observations are labelled as such, and one non-deterministic result was excluded rather than reported as a confirmed defect.

3 Severity definitions

SEVERITY	DEFINITION
CRITICAL	Blocks a core user journey or the launch for all users.
HIGH	Breaks a primary journey or affects every visitor, or creates SEO or legal risk.
MEDIUM	Degrades the experience for some content, users or environments.
LOW	Polish, consistency or best-practice issue.

Type shows whether a defect is a **template defect** (code that ships with the starter) or a **content-handling gap** (the code trusts CMS content that can legitimately arrive empty, long or broken).

4 Test environment

PROFILE	ENGINE	OS	VIEWPORT	TYPE
Desktop Chrome	Chromium 153	macOS	1280×720, plus a 360/768/1280 sweep	Real engine
Desktop Safari	WebKit 26.6 (Playwright build)	macOS	1280×720, plus the sweep	WebKit engine
Desktop Firefox	Firefox 155	macOS	1280×720, plus the sweep	Real engine
iPhone 14	WebKit	iOS profile	390×664 @3x, touch	Emulated
Pixel 7	Chromium	Android profile	412×839 @2.6x, touch	Emulated
iPad (gen 7)	WebKit	iPadOS profile	810×1080 @2x, touch	Emulated

- Desktop Safari runs Playwright's WebKit build, not Apple's Safari binary.
- Each finding states the environment it came from. On a client project, emulated-only findings would be confirmed on real iPhone, iPad and Android devices (all three are available).

5 Findings overview

ID	FINDING	SEVERITY	TYPE
CMS-01	Page routing — unknown URLs return HTTP 200 with a public “Create Page” Studio prompt (soft 404)	HIGH	Template defect
CMS-02	Rich text — an empty field renders a stray “0” on the page	MEDIUM	Template defect
CMS-03	Layout — unbreakable words and pasted URLs cause horizontal scrolling	MEDIUM	Content-handling gap
CMS-04	Localisation — Arabic (RTL) content renders left-to-right, with misordered punctuation and numbers	MEDIUM	Content-handling gap
CMS-05	Media — a deleted image asset leaves a broken image and a large empty gap	MEDIUM	Content-handling gap
CMS-06	Call to action — a button whose linked page was deleted becomes dead plain text	MEDIUM	Template defect
CMS-07	Call to action — every CTA image gets <code>alt="Demo image"</code> regardless of content	MEDIUM	Template defect
CMS-09	Routing — duplicate slugs make one published post unreachable	MEDIUM	Content-handling gap
CMS-11	Sitemap — scheme-less URLs, a “/null” entry and duplicate entries	MEDIUM	Template defect
CMS-12	Post cards — card links have no accessible name (WCAG 2.4.4 and 4.1.2)	MEDIUM	Template defect

6 Detailed findings

CMS-01 **HIGH**

Template defect

Page routing — unknown URLs return HTTP 200 with a public “Create Page” Studio prompt (soft 404)

IMPACT Every mistyped or outdated URL is indexable as a real page, and public visitors see an internal editor prompt that links into the Studio.

ENVIRONMENT All browsers; server response (also reproduced with `curl`).

ROOT CAUSE `app/[slug]/page.tsx` renders `<PageOnboarding/>` instead of calling `notFound()`.

STEPS TO REPRODUCE

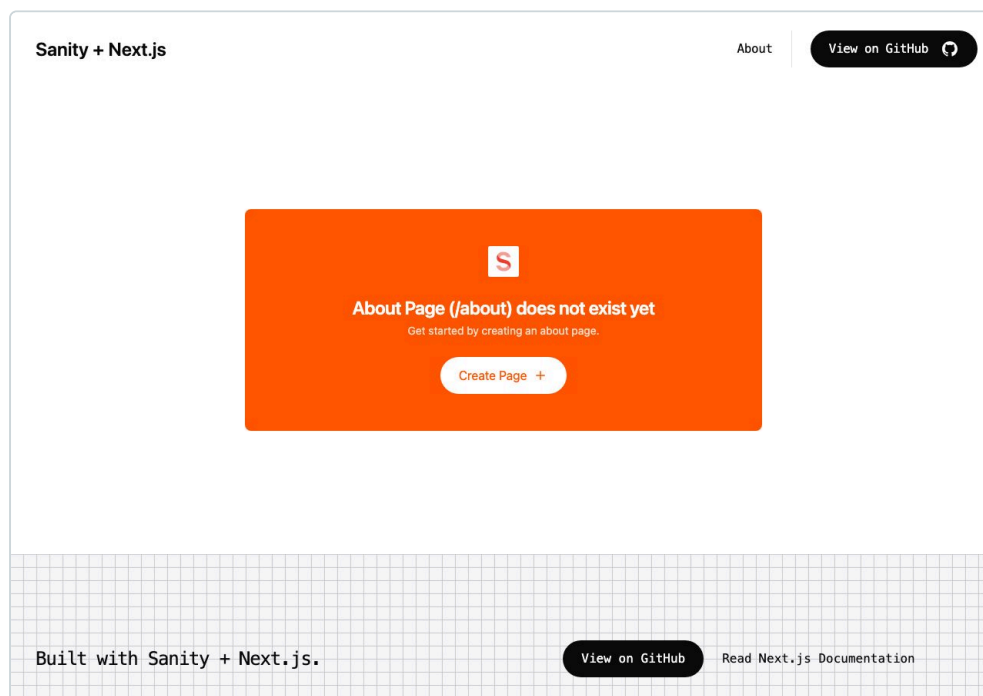
1. Open `/this-page-does-not-exist`.
2. Check the response status in DevTools or with `curl -I`.

EXPECTED

The server returns a 404 status with a not-found page (or the page is marked `noindex`).

ACTUAL

The server returns 200 OK with an orange card reading “About Page (/about) does not exist yet. Get started by creating an about page. Create Page”, which names `/about` whatever the URL and links to the Studio; unknown `/posts/...` URLs correctly return 404.



`/this-page-does-not-exist` served with HTTP 200 and the Studio onboarding card.
evidence/CMS-01-unknown-url-200.png

RECOMMENDATION Call `notFound()` on the public site when `page?._id` is missing, and show the onboarding card only in Draft Mode or development.

CMS-02 **MEDIUM**

Template defect

Rich text — an empty field renders a stray “0” on the page

IMPACT A visible glitch appears whenever an editor leaves a post body or an info section empty.

ENVIRONMENT All browsers.

ROOT CAUSE `{post.content?.length && ...}` in `app/posts/[slug]/page.tsx:97` and `InfoSection.tsx:25`: React renders the number `0`. Triggered by content.

STEPS TO REPRODUCE

1. Create a post with an empty **Content** field (`[]`), or a page-builder **Info section** with empty content.
2. Open `/posts/qa-empty-fields` or `/qa-landing`.

EXPECTED

Nothing is rendered for empty content.

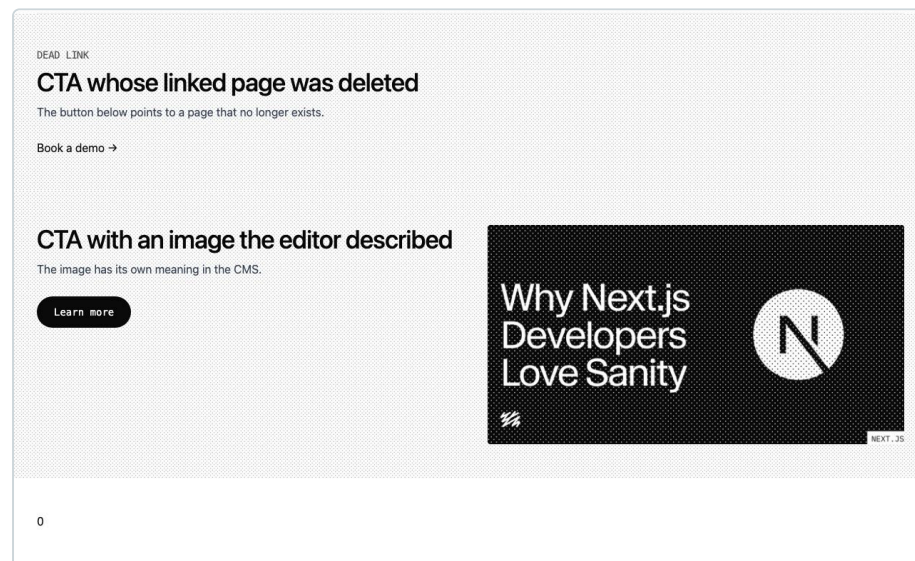
ACTUAL

A lone “0” appears where the body would be, on the post page and below the CTAs on the landing page.



0

Post page: the stray “0” where the body should be.
`evidence/CMS-02-stray-zero.png`



Landing page (cropped): the same “0” below the CTAs.
`evidence/CMS-06-dead-cta.png` (cropped)

RECOMMENDATION Use `content?.length > 0 && ...` or `!!content?.length && ...`.

CMS-03 **MEDIUM** *High on mobile*

Content-handling gap

Layout — unbreakable words and pasted URLs cause horizontal scrolling

IMPACT The whole page scrolls sideways, the title is clipped and the header button is pushed off-screen; mobile visitors are hit hardest.

ENVIRONMENT Chromium, WebKit and Firefox desktop engines at 360, 768 and 1280 px. The post page overflows at all three widths (for example `scrollWidth` 1491 in a 1280 px viewport); the home page overflows at 360 px because of the post card. Emulated phones show the same.

ROOT CAUSE No `overflow-wrap: anywhere` or `break-words` on CMS-driven headings, cards or body text.

STEPS TO REPRODUCE

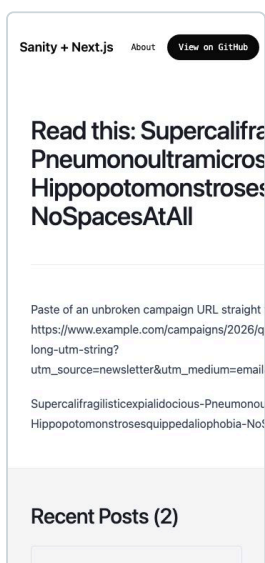
1. Publish a post whose title contains a long unbroken word, or whose body contains a long campaign URL (entry `qa-unbreakable-word`).
2. Open `/posts/qa-unbreakable-word` at 360 px.
3. Open the home page at 360 px.

EXPECTED

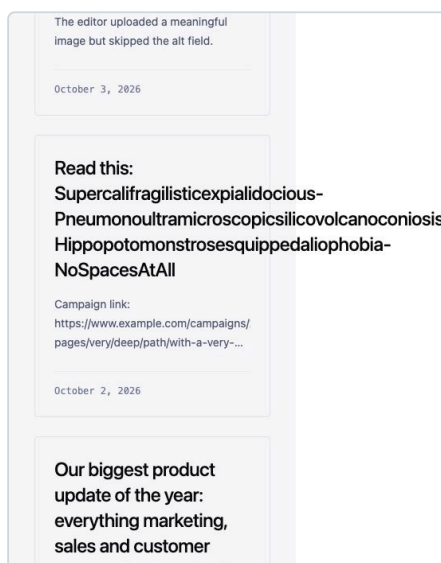
Text wraps inside its container and the page does not scroll horizontally.

ACTUAL

The `h1` (`text-7xl`) and the body paragraph extend past the viewport, and the page scrolls horizontally.



Post page at 360 px: title and URL run off-screen.
evidence/CMS-03-unbreakable-360.png



Home page at 360 px (cropped): the post card overflows.
evidence/CMS-03-home-card-360.png (cropped)

RECOMMENDATION Add `break-words` or `[overflow-wrap:anywhere]` to the `h1`, the card `h3` and `.prose`, and consider `hyphens: auto`.

CMS-04 **MEDIUM** High for MENA or multilingual clients

Content-handling gap

Localisation — Arabic (RTL) content renders left-to-right, with misordered punctuation and numbers

IMPACT Arabic text reads incorrectly and % , . and 20 jump to the wrong side; for any MENA or multilingual client this rises to High.

ENVIRONMENT All engines.

ROOT CAUSE `<html lang="en">` for all content, with no `dir` or `lang` per document or block.

STEPS TO REPRODUCE

1. Publish a post with an Arabic title, excerpt and body (`qa-arabic-rtl`).
2. Open `/posts/qa-arabic-rtl` .

EXPECTED

The article is right-aligned and reads right-to-left (`dir="rtl"` and `lang="ar"` on the article, or `dir="auto"` on text blocks).

ACTUAL

Headings and paragraphs are left-aligned and LTR: "15%." shows as "%." at the start of the next line, and the mixed line 31/12 في العرض ينتهي في %20 use code SAVE20 — renders as "SAVE20 20 للحصول...", with the numbers reordered.



Arabic post rendered left-to-right with misplaced punctuation.
evidence/CMS-04-arabic-ltr.png

RECOMMENDATION Add a `language` field (or detect the script) and set `dir` and `lang` on the article; use `dir="auto"` on Portable Text blocks.

CMS-05

MEDIUM

Content-handling gap

Media — a deleted image asset leaves a broken image and a large empty gap

IMPACT Pages look broken after a routine media-library clean-up.

ENVIRONMENT All engines.

ROOT CAUSE The cover image renders whenever `coverImage` exists, even if its asset no longer does.

STEPS TO REPRODUCE

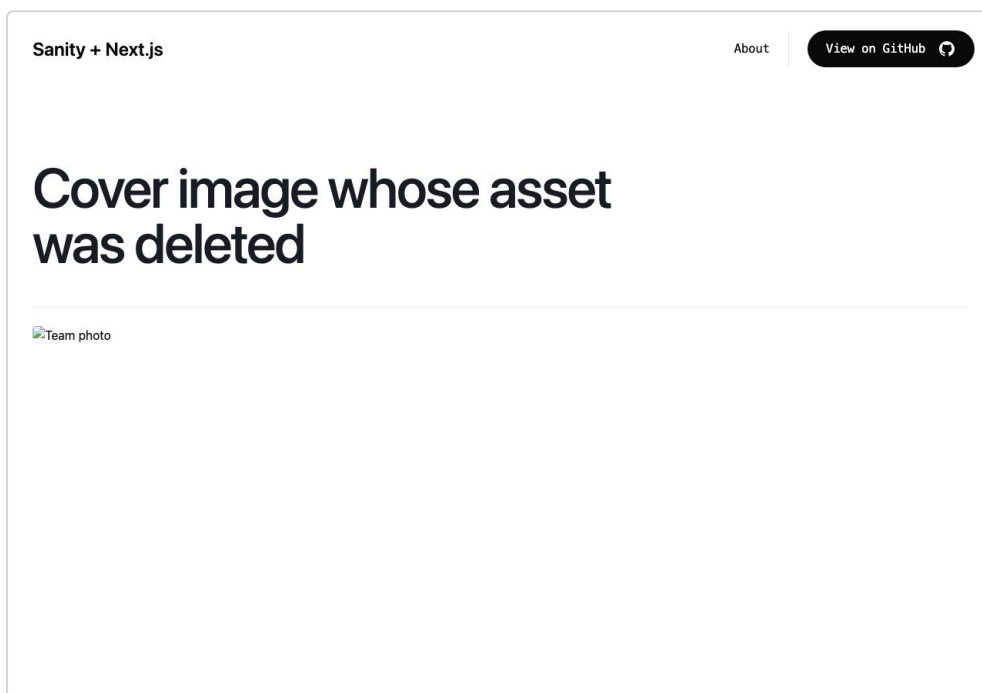
1. Publish a post with a cover image.
2. Delete the image asset (entry `qa-broken-image`, a weak reference).
3. Open `/posts/qa-broken-image`.

EXPECTED

No image, or a branded placeholder, is shown.

ACTUAL

A broken-image icon with the alt text "Team photo" is followed by roughly 540 px of empty reserved space, and the CDN request fails with a 4xx.



Broken image followed by a large empty gap.
`evidence/CMS-05-broken-image.png`

RECOMMENDATION Dereference the asset in GROQ (`coverImage{..., asset→}`) and render the image only when `asset→url` is defined.

CMS-06 **MEDIUM**

Template defect

Call to action — a button whose linked page was deleted becomes dead plain text

IMPACT A conversion path breaks silently; because it no longer looks like a button, editors are unlikely to notice.

ENVIRONMENT All engines.

ROOT CAUSE `ResolvedLink` returns bare `children` when the link cannot be resolved, dropping the button styling.

STEPS TO REPRODUCE

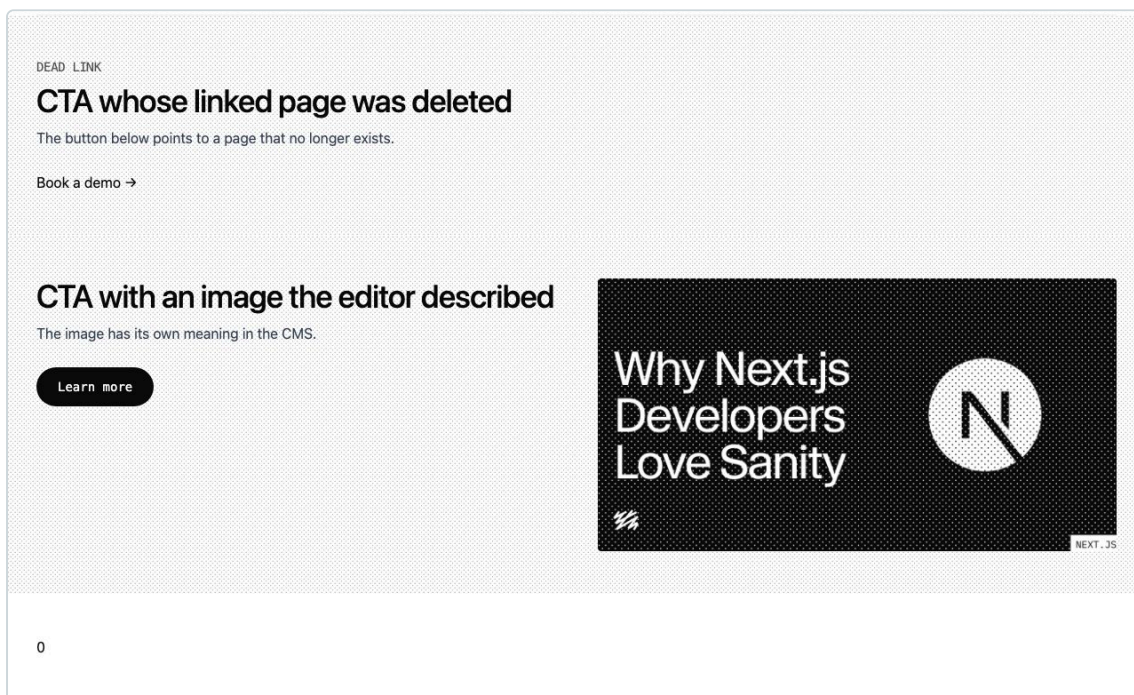
1. Create a Call to Action whose button links to a page.
2. Delete that page (`qa-landing` , first CTA).
3. Open `/qa-landing` .

EXPECTED

The button is hidden (or flagged in preview) and never shown as dead text.

ACTUAL

"Book a demo →" renders as unstyled, non-clickable text below the CTA heading.



Landing page (cropped): "Book a demo →" rendered as plain text.
evidence/CMS-06-dead-cta.png (cropped)

RECOMMENDATION In `Cta.tsx` , render the button only when `linkResolver(button.link)` returns a string, and show a warning badge in Draft Mode.

CMS-07 **MEDIUM**

Template defect

Call to action — every CTA image gets `alt="Demo image"` regardless of content

IMPACT Screen readers and image search receive the same wrong description on every landing page.

ENVIRONMENT All engines (DOM check).

ROOT CAUSE `alt="Demo image"` is hard-coded in `Cta.tsx`, and the CTA image schema has no alt field.

STEPS TO REPRODUCE

1. Add a CTA with an image (`qa-landing`, second CTA).
2. Inspect the rendered image.

EXPECTED

The alt text comes from the CMS.

ACTUAL

The image renders as `<img alt="Demo image">`.

EVIDENCE DOM inspection; no screenshot required.

RECOMMENDATION Add an `alt` field to the CTA image (required when an image is set) and render it.

CMS-09 **MEDIUM**

Content-handling gap

Routing — duplicate slugs make one published post unreachable

IMPACT One article silently disappears: its card opens a different post.

ENVIRONMENT All engines.

ROOT CAUSE Slug uniqueness is enforced only in the Studio UI; API, import and migration writes bypass it.

STEPS TO REPRODUCE

1. Publish two posts with the slug `qa-duplicate-slug`.
2. On the home page, click each of the two cards.

EXPECTED

Each post has a unique URL, or a build or CI check fails on duplicates.

ACTUAL

Both cards link to `/posts/qa-duplicate-slug` and both open "Duplicate slug A"; post B cannot be reached, and the sitemap lists the URL twice.

EVIDENCE Link targets checked in the DOM; the duplicate also appears in the sitemap (CMS-11).

RECOMMENDATION Add a uniqueness check to import and migration scripts, plus a CI query: `count(*[_type="post" && slug.current==^.slug.current]) > 1`.

CMS-11 **MEDIUM** SEO

Template defect

Sitemap — scheme-less URLs, a `"/null"` entry and duplicate entries

IMPACT Search engines reject or ignore sitemap entries that have no scheme.

ENVIRONMENT Server output (`/sitemap.xml`).

ROOT CAUSE `url: domain` uses the raw `Host` header (`localhost:3000` , no `https://`). In `sitemapData` , `_type = "page" || _type = "post" && defined(slug.current)` includes pages without a slug because of operator precedence, which produces `/null` .

STEPS TO REPRODUCE

1. Open `/sitemap.xml` .

EXPECTED

Every entry is an absolute `https://` URL, with no null slugs and no duplicates.

ACTUAL

The sitemap contains

```
<loc>localhost:3000/about</loc> ,  
<loc>localhost:3000/null</loc> and  
.../posts/qa-duplicate-slug twice.
```

```
<loc>localhost:3000</loc>  
<loc>localhost:3000/about</loc>  
<loc>localhost:3000/qa-landing</loc>  
<loc>localhost:3000/null</loc>           ← page without a slug  
...  
<loc>localhost:3000/posts/qa-duplicate-slug</loc>  
<loc>localhost:3000/posts/qa-duplicate-slug</loc>   ← duplicate
```

Excerpt of the `<loc>` entries (19 in total) · [evidence/CMS-11-sitemap.xml](#)

RECOMMENDATION Build URLs from a configured site URL, wrap the filter in parentheses (`( _type="page" || _type="post") && defined(slug.current)`) and de-duplicate the entries.

CMS-12 **MEDIUM** Accessibility

Template defect

Post cards — card links have no accessible name (WCAG 2.4.4 and 4.1.2)

IMPACT Screen-reader users hear “link” 15 times on the home page, with no destination.

ENVIRONMENT axe-core 4.13 on Chromium (desktop) and emulated iPhone; serious impact, 15 nodes on / .

ROOT CAUSE The card’s `<Link>` wraps only an empty `absolute inset-0` span; the title sits outside the link.

STEPS TO REPRODUCE

1. Open / and wait for the post list to load.
2. Run axe, or tab through the cards with VoiceOver.

EXPECTED

Each card link is announced with the post title.

ACTUAL

axe reports `link-name` violations: “Element does not have text that is visible to screen readers”.

```
{ "id": "link-name", "impact": "serious", "nodes": 15,
  "help": "Links must have discernible text",
  "sample": [ "a[href$=\"qa-empty-fields\"]" ] }
```

axe result for the home page · [evidence/CMS-12-axe-home.json](#)

RECOMMENDATION Place the title inside the link, or add `aria-label={title}` to the overlay link.

7 Minor findings

Low-severity findings, each confirmed and encoded in the regression suite.

ID	FINDING	TYPE	ENVIRONMENT	SEVERITY
CMS-08	A cover image without CMS alt text silently becomes <code>alt=""</code> (decorative) instead of being flagged.	Content-handling gap — Studio's "alt required" rule does not apply to API writes	All	LOW
CMS-10	A post with an empty excerpt gets no meta description; there is no fallback to the site description or a body extract.	Template defect	All	LOW
CMS-13	Colour contrast: post dates (<code>#727892</code> on white) at 4.35:1, and orange inline links (<code>#ff5500</code> on white) at 3.2:1; 4.5:1 is required.	Template defect (design tokens)	axe, Chromium; <code>evidence/CMS-13-axe-*.json</code>	LOW
CMS-14	The home hero overflows by 7 px at 768 px in the WebKit engine only (the starter's demo hero, not CMS content).	Template defect	WebKit desktop engine; <code>evidence/CMS-14-webkit-home-768.png</code>	LOW

EMULATION-ONLY OBSERVATION (NEEDS A REAL DEVICE)

On emulated Pixel 7, the Draft Mode toast's **Disable** button could not be tapped on the home page: Playwright reported the page container intercepting the tap. The same flow works on emulated iPhone and iPad, and on Pixel 7 from a page without overflow, so it is most likely a side effect of CMS-03 (the 412 px home page scrolls sideways). It is not reported as a separate defect until it is confirmed on a real Android phone.

8 Verified as working

Behaviour that was tested and found correct.

Draft Mode	Drafts are hidden on the public site (404, and not listed), and published posts show the published version, not unpublished changes. With a valid preview secret, the draft-only post, the unpublished v2 and the draft in the listing all appear, the “Draft Mode Enabled” toast shows, and Disable returns to published content. An invalid preview secret is rejected. Passes in all six profiles.
Output escaping	<code><script></code> , <code></code> and <code>&amp;</code> in titles and excerpts render as text; no script runs and no dialog opens.
Deleted author	Handled cleanly, with no dangling “By” line.
Unknown posts and links	Unknown post URLs return 404, including an unknown Arabic slug. The internal link crawl (up to 60 URLs from home, about, landing and a post) found no 4xx or 5xx links.
Long headlines with spaces	A 230-character headline with normal spaces wraps correctly at every breakpoint, and an empty info-section heading leaves no empty heading element.

9 Key QA insights for CMS-driven applications

Valid in the CMS is not valid on the page

Studio validation (required alt text, unique slugs, required link targets) runs only in the editor. Imports, migrations, API scripts and drafts bypass it, and several findings (CMS-03, 04, 05, 08 and 09) appear only once content arrives that way.

References rot over time

A hardcoded page cannot link to a page that does not exist; a CMS page can, once someone deletes the target (CMS-05 image, CMS-06 CTA). These defects appear weeks after launch, so regression tests must seed broken references deliberately.

Empty and missing behave differently

`[]`, `""` and an absent field each take a different code path: `[]` produced CMS-02's "0", while `""` and an absent field were handled. Testing only "filled" and "absent" misses it.

One component, many pages

A single card component carries CMS-03, CMS-09 and CMS-12 to every listing (home and "More posts"), so one CMS defect multiplies across pages.

Preview is a second product

Draft Mode has its own rendering path (perspective, toast, live updates) and needs its own test pass. Here it worked, but its exit control was the one item affected by an unrelated layout defect on emulated mobile.

10 AI-assisted workflow and validation controls

USED FOR

AI accelerated adversarial-content generation, initial Playwright scaffolding, axe/console parsing and structured DOM/sitemap analysis. No finding was accepted from AI or automation output alone; every reported defect was independently re-run and verified.

Validation controls and corrections

- 1. Overflow detector missed a clear overflow.** The AI-written check compared element bounding boxes with the viewport and reported “no offending elements” while the page clearly scrolled sideways: long unbroken text overflows its box without the box itself growing. A second check (`scrollWidth > clientWidth` with `overflow: visible`) was added, and CMS-03 was confirmed visually.
- 2. First axe run reported a clean home page.** The scan ran before the post list streamed in (React Suspense), so 15 `Link-name` violations were missed. The suite now waits for network idle, and CMS-12 is real.
- 3. Discarded — “uppercase slugs return 200”.** One run showed `/posts/QA-LONG-TITLE` returning 200. Repeating it gave 200 or 404 depending on the ISR cache state (GROQ itself is case-sensitive). The result is not deterministic, so it is not reported as a defect; it is noted for a follow-up on the deployed host.
- 4. Over-reported — Pixel “Disable” failure.** The AI drafted it as a separate mobile defect. On manual re-check it occurs only on the overflowing page, so it is downgraded to an emulation-only observation linked to CMS-03.

11 Regression suite

See `README.md` for setup. `npm test` runs the whole matrix; `npx playwright test --project=desktop-chrome` runs a single profile, and `npm run report` opens the HTML report.

Confirmed defects call `knownBug('CMS-xx ...')` and assert the correct behaviour, so they show as expected failures. When a fix lands, the matching test reports an unexpected pass: remove its marker, and it becomes a permanent regression check.